Random experiment: Roll a die, HT

outcome: → Result of random experiment roll a die - HT

events: is acollection of outcomes

sample space: set of all possible outcomes

3 Axioms

1- $P(A) \geq 0$

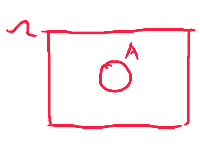
2- $P(\Omega) = P(S) = 1$

3 - $A_1, A_2 \cdots A_n$ are disjoint events

$P(A_1 \cup A_2 \cdots \cup A_n) = P(A_1) + P(A_2) \cdots P(A_n)$

6 Rules

1-) $P(A) = 1 - P(A^c)$



$A \cup A^c = \Omega$

$P(A \cup A^c) = 1$ $\quad \underline{P(A)} + P(A^c) = 1$

2-) $P(\emptyset) = 0$

$P(\emptyset) = P(\Omega^c) \rightarrow 1 - P(\Omega) = 0$

3-) $P(A) \leq 1$

$P(A) = 1 - \underbrace{P(A^c)}_{\geq 0} \rightarrow P(A) \leq 1$

4-) $P(A - B) = P(A) - P(A \cap B)$

$A = (A \cap B) \cup (A - B)$ $\quad P(A) = P(A \cap B) + P(A - B)$

$P(A - B) = P(A) - P(A \cap B)$

5-) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

6-) $A \subset B \overset{\text{if}}{\Longrightarrow} P(A) \leq P(B)$

Axioms are important

↳ Rules of prob. ar obtained using these axioms.

↳ Any function that .... these axioms define a prob. measure

Calculating prob.

↳ Equially likely outcomes.

-Rolling a die

$\Omega = \{1,2,3,4,5,6\} \rightarrow P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{|\Omega|}$

Rolling a fair die and get an odd numbers?

$P(odd) = \overset{\{1,5,3\}}{\{1\} + \{2\} + \{3\}} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$

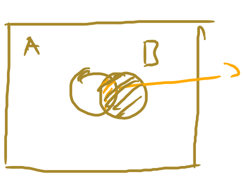
Conditional probability (koşullu olasılık) condition → koşul

P(rain)=? $\quad$ P(rain) = 0.4

→ Clouds in the sky → P(rain) = 0.85

$P(rain \underset{\text{given}}{|} clouds) = 0.75$

Definition: The conditional probability $P(A|B)$ prob. that A occured given B has occured is: $P(A/B) = \frac{P(A \cap B)}{P(B)}$



$\rightarrow \frac{P(A \cap B)}{P(B)}$ $\quad P(B) \neq 0$

Ex: Roll a fair die

prob. of getting an even number?

given the outcome is less then or equal to 3 → P(even / outcome ≤ 3)

$\Omega = 1,2,3$ $\quad P(even) = \{2\} \rightarrow \frac{1}{3}$

ex/ Flight we know that 90% of the flight departs on time

80% of the flights arrive on time

70% of the flights both arrive and depart on time

a) We know that flight departed on time. Prop. that it will arrive on time?

$P(A \cap B) = 70\%$ $\quad P(A) = 90\%$ $\quad P(B/A) = \frac{70}{90} = \frac{7}{9}$

b) " " " " arrived , " " " " " depart " "?

$P(B) = 80\% \rightarrow P(A/B) = \frac{70}{80} = \frac{7}{8}$

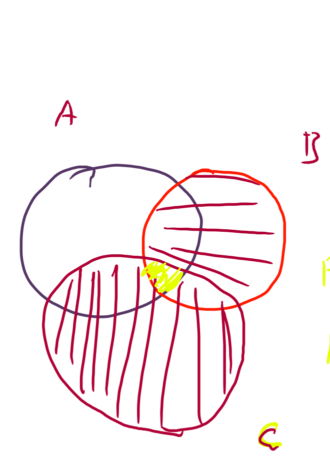
if there are more than 2 events?

P(rain / cloudy, windy) = ?

Chain rule

$P(A \cap B) = P(A|B) \cdot P(B)$ → cond. prob.

$P(A \cap B \cap C) = P(C) \cdot P(B/C) \cdot P(A/B, C)$



$P(A/B, C)$

$P(A \cap B \cap C)$

independence of events

definition → 2 events A and B are independent iff : ⟺

$P(A/B) = \underset{\sim}{P(A)}$

it is independent of B

$P(A \cap B) = P(A) \cdot P(B)$

A: rain → yes/No

B: flip a coin → H/T

$P(\underset{A}{rain} / \underset{B}{T})$

$P(A|B) = \frac{P(A \cap B)}{P(B)} \rightarrow P(A) \cdot P(B) = P(A \cap B)$

Bağımsız olaylar ayrık olmak zorunda değil

independent events are not $\underline{\text{disjoint}}$

Ex/: I roll a die and observe number x. Let A be an event that x is

gözlemek

an even number, and let B be an event that x > 4.

Are A and B independent?

$A \{2,4,6\}$ $\quad B \{5,6\}$ $\quad P(A) = \frac{1}{2}$ $\quad P(B) = \frac{1}{3}$

$P(A \cap B) = \frac{1}{6}$ $\quad \frac{1}{6} \overset{?}{=} \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$ → Events A an B independent

$P(A \cap B) \overset{?}{=} P(A) \cdot P(B)$

Flight example

$P(A) = 80\%$ $\quad P(D) = 90\%$ $\quad P(A \cap D) = 70\%$ $\quad$ Are A and D independent?

$\frac{4}{5} \cdot \frac{9}{10} \overset{?}{=} \frac{7}{10} \rightarrow \frac{18}{25} \neq \frac{7}{10}$ $\quad$ events A and B are not independent

or $P(A/D) \overset{?}{=} P(A) \rightarrow \frac{70\%}{90\%} \neq 80\%$ $\quad$ " " " " " " "

or $P(D/A) \overset{?}{=} P(D) \rightarrow \frac{70\%}{80\%} \neq 90\%$

independance of 3 events

$P(A \cap B) = P(A) \cdot P(B)$ $\quad P(A \cap C) = P(C) \cdot P(A)$ $\quad P(B \cap C) = P(B) \cdot P(C)$

$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$ ise A, B and C are independent

Conditional independent

Two event A and B are Conditionally independent

given C iff

$P(A \cap B / C) = P(B/C) \cdot P(A/C)$

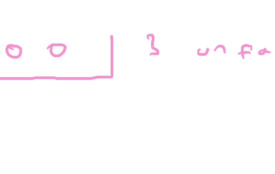
law of total probability

Let $B_1, B_2, \ldots, B_n$ be a partition → $B_1 \cup B_2 \ldots B_n = \Omega$

of Ω and $P(B_i) > 0$ $\quad B_1 \cap B_2 = \emptyset$

For any event A we have

$P(A) = \sum_i P(A \cap B_i) = \sum P(A/B_i) \cdot P(B_i)$



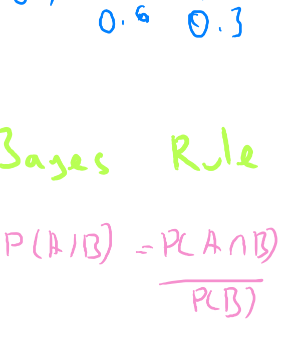
$(A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3) = P(A)$

ex/ [0 0 0] 3 unfair coins we know that - for coin 1 P(H) = 0.9

- " " 2 P(H) = 0.6

- " " 3 P(H) = 0.3

we draw a coin randomly and we flip P(H)=?



$P(A) = \sum_i P(A \cap B_i) = \sum P(A/B_i) \cdot P(B_i)$

$= 0.9 \cdot \frac{1}{3} + 0.6 \cdot \frac{1}{3} + 0.3 \cdot \frac{1}{3} = 0.6$

Bayes Rule

$P(A|B) = \frac{P(A \cap B)}{P(B)}$ $\quad P(B/A) = \frac{P(A \cap B)}{P(A)}$

$\rightarrow P(A/B) \cdot P(B) = P(B/A) \cdot P(A)$ $\rightarrow$ $P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$ ← Bayes rule ☆

ex/ if we get H, what is the prob. we draw the first coin?

$P(C_1/H) = \frac{\overset{9/10}{P(H/C_1)} \cdot \overset{1/3}{P(C_1)}}{\underset{3/5}{P(H)}} \rightarrow \frac{1}{2}$